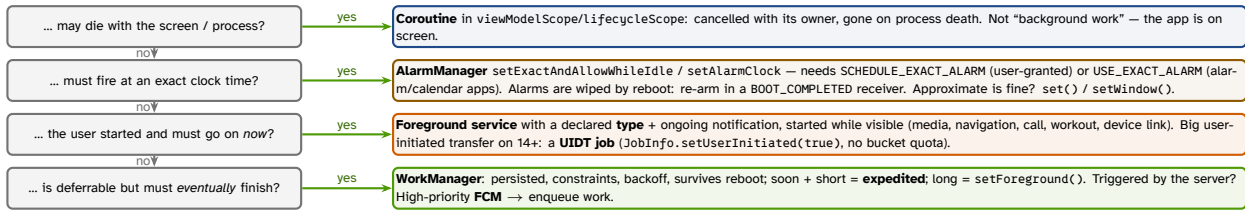


In one line: Android runs your code in the background only on its terms: a **coroutine** dies with its owner, **WorkManager** guarantees deferrable work (persisted, survives reboot, runs when quota and constraints allow), a **foreground service** is visible work the user started, and **AlarmManager** is for a wall-clock moment. Every release since 6.0 (Doze) has narrowed the rest: standby-bucket quotas, background-start bans, typed and time-limited services.

“I need to run work that ...” — ask in this order



None of these? Reconsider: iOS-style “keep running in the background” does not exist. WorkManager sits on JobScheduler — call JobScheduler directly only for UIDT jobs.

WorkManager internals

- enqueue writes a WorkSpec row to WorkManager's own **Room** DB, then schedules it with JobScheduler (and runs it in-process at once if constraints already hold). Reboot, update, force-stop: rows re-registered (boot receiver, next launch).
- One run of a worker $\approx$ one job: the system stops it after **10 min** (CoroutineWorker is cancelled, work is retried). Longer: setForeground() → SystemForegroundService — declare it in the manifest with your FGS type.
- Numbers: backoff default **EXPONENTIAL 30 s**, min 10 s, max 5 h; periodic $\geq$ **15 min**, flex $\geq$ 5 min; Data $\leq$ 10 KB — pass URIs/ids.
- enqueueUniqueWork(name, KEEP|REPLACE|APPEND|APPEND_OR_REPLACE, r) (periodic: KEEP|UPDATE). beginWith(listOf(a, b)).then(c): outputs merged by an InputMerger; a failed parent fails the chain.
- Expedited:** setExpedited(RUN_AS_NON_EXPEDITED_WORK_REQUEST | DROP_WORK_REQUEST); quota by standby bucket, none while in foreground; below Android 12 it runs as an FGS via getForegroundInfo() — missing it = crash.

Example — a resilient upload

```

@HiltWorker class UploadWorker @AssistedInject constructor(
    @Assisted ctx: Context, @Assisted p: WorkerParameters, val api: Api
) : CoroutineWorker(ctx, p) {
    override suspend fun doWork(): Result = try {
        setForeground(info()) // may outlive 10 min
        api.upload(inputData.getString("uri")!!); Result.success()
    } catch (e: IOException) {
        if (runAttemptCount < 5) Result.retry() else Result.failure()
    }
    override suspend fun getForegroundInfo() = info() // expedited < 12
    private fun info() = ForegroundInfo(42, notification(),
        ServiceInfo.FOREGROUND_SERVICE_TYPE_DATA_SYNC)
    WorkManager.getInstance(ctx).enqueueUniqueWork("upload",
        ExistingWorkPolicy.KEEP, OneTimeWorkRequestBuilder<UploadWorker>()
            .setConstraints(Constraints.Builder()
                .setRequiredNetworkType(NetworkType.UNMETERED).build())
            .setInputData(workDataOf("uri" to uri)).build())
  }

```

Traps

- ! “WorkManager for a 30-min download the user tapped” — FGS or UIDT job.
- ! Tested on a plugged-in phone — force Doze and a rare bucket.
- ! Thread/coroutine in a BroadcastReceiver or Application — the process can be killed the moment no component is active.

Remember: dies with screen → coroutine · clock → alarm · visible now → typed FGS · must finish → WorkManager.

Foreground services — the rules by version

- 12+:** no FGS start from the background → ForegroundServiceStartNotAllowedException. Exemptions: high-priority FCM, exact alarm, notification/widget tap, some BOOT_COMPLETED starts, SYSTEM_ALERT_WINDOW (15+: only with a visible overlay).
- 14+:** android:foregroundServiceType + FOREGROUND_SERVICE_<TYPE> permission, else MissingForegroundServiceTypeException; declared in Play Console. Camera/mic/location types can't start in background even when exempt. shortService: ~3 min.
- 15+:** dataSync and mediaProcessing get **6 h per 24 h** (each), then Service.onTimeout(int, int) — not stopped = crash; foreground resets it. BOOT_COMPLETED may not start dataSync, mediaPlayback, mediaProjection, phoneCall.

Doze & App Standby Buckets

Doze (unplugged, still, screen off): no network, wakelocks ignored, jobs/syncs/alarms wait for **maintenance windows**; ...AllowWhileIdle alarms $\leq$ once per **9 min** per app; high-priority FCM gets brief network. **Buckets** (usage, ML-predicted) cap work:

Bucket	Regular jobs	Expedited	Alarms
Active	20 min / 60 min	30 min / 24 h	no limit
Working set	10 min / 4 h	15 min / 24 h	10 / h
Frequent	10 min / 12 h	10 min / 24 h	2 / h
Rare (no network)	10 min / 24 h	10 min / 24 h	1 / h
Restricted (12+)	1 run / day, 10 min	5 min / 24 h	1 / day

16: jobs started while visible or running beside an FGS now count against the quota. Test: am set-standby-bucket <pkg> rare · dumpsys deviceidle force-idle.

Exact alarms & broadcasts

- 12:** exact needs SCHEDULE_EXACT_ALARM; **13:** or USE_EXACT_ALARM (auto-granted, not revocable, policy-limited); **14:** the former is **denied by default** on new installs — canScheduleExactAlarms(). Inexact set(): within 1 h.
- Manifest receivers get no implicit broadcasts (8.0+) bar a short list (BOOT_COMPLETED, ...). onReceive runs on main, <10 s; goAsync() buys time, not a lifetime — hand off to WorkManager.